Band-1 Hadamard H_508 Ingest + Miyamoto Construction of H_2036

Finite constructions of orders 508 and 2036 ONLY; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_2036_method_triage`; `2026-09-14_Band1_Hadamard_2004_lit_watch`.
Lab. `/workspace/experiments/band1/hadamard_508_to_2036_miyamoto/`.
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo reports **finite constructions** of Hadamard matrices of orders **508** and **2036** on this box, verified by $\max |HH^\top - nI| = 0$. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. `clay_closed = false`.

Abstract

Prior Band-1 triage showed that order **2036** is Miyamoto-eligible as an existence reduction to literature H_{508} ($q = 509$ prime, $509 \equiv 1 \pmod 4$), but no H_{508} payload was on box. This increment constructs H_{508} via the Goethals–Seidel array from a published skew supplementary difference set of order 127 (SageMath/GPL SDS data; parameters 4-$\{127; 63, 57, 57, 57; 107\}$), verifies $HH^\top = 508I$, then applies the Miyamoto (1991) construction with the Paley symmetric conference matrix of order 510 to produce H_{2036}, verified $HH^\top = 2036I$. Unit tests **15/15 PASS**. Wall time ≈ 4.4s. **Claim level:** finite constructions of orders 508 and 2036 **only**. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 508; order 2036; Miyamoto construction; Goethals–Seidel; SDS; Band-1.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment).

1. An explicit ± 1 matrix of order **508** constructed on box and verified.
2. An explicit ± 1 matrix of order **2036** constructed on box via Miyamoto and verified.

Not claimed. The Hadamard conjecture for all admissible orders.

2. Environment and honesty gates

Item	Status
SageMath CLI / `import sage`	**ABSENT** on box
Construction path	Pure NumPy reimplementation of published SDS + GS + Miyamoto
GATE-04 collection	**LOCKED** (no invented J)
`clay_closed`	**false**
Full conjecture	**OPEN** (Band-3)

3. Method triage for order 508

Method	Result
Paley I ($q = 507$)	**FAIL** ($507 = 3 \cdot 13^2$ not a prime power)
Paley II ($q = 253$)	**FAIL** ($253 = 11 \cdot 23$ not a prime power)

Method	Result
Kronecker nontrivial	**FAIL** ($508 = 4 \cdot 127$ but 127 not Hadamard-admissible)
Miyamoto for 508 ($q = 127$)	**INELIGIBLE** ($127 \equiv 3 \pmod 4$); classical theorem needs $q \equiv 1 \pmod 4$)
Goethals–Seidel from SDS order 127	**PASS** (skew SDS published; verified on box)

4. Construction of H_{508}

Input. Skew spin-type SDS of order 127 (SageMath `skew_spin_goethals_seidel_difference_family` compact data):

- $H = \{1, 2, 4, 8, 16, 32, 64\} \subset \mathbb{Z}/127\mathbb{Z}^\times$
- representatives and multiplier yielding SDS sizes $(63, 57, 57, 57)$ with $\lambda = 107$
- check: $\sum k_i = 127 + \lambda$ and constant difference count λ

Array. First rows $a_i[j] = -1$ iff $j \in S_i$; circulant blocks A, B, C, D; Goethals–Seidel array of order $4 \cdot 127 = 508$.

Verification. Entries in $\{\pm 1\}$; $\max |HH^\top - 508I| = 0$ (blocked row products, int32).

Source tag. `skew_spin_compact_127`.

5. Miyamoto construction of H_{2036}

Theorem (Miyamoto 1991). If q is a prime power with $q \equiv 1 \pmod 4$ and a Hadamard matrix of order $q - 1$ exists, then a Hadamard matrix of order $4q$ exists.

Here. $q = 509$ prime, $509 \equiv 1 \pmod 4$, H_{508} from Section 4; Paley symmetric conference matrix of order 510 ($q = 509 \equiv 1 \pmod 4$).

Verification. Shape $(2036, 2036)$; entries ± 1; $\max |HH^\top - 2036I| = 0$.

Claim. Finite construction of order **2036** only — **not** the full conjecture.

6. Unit tests

Test	Result
H_{20} Paley I control	PASS
H_4 Sylvester	PASS
H_{12} Paley I	PASS
508 admissible	PASS
2036 divisible by 4	PASS

Test	Result
509 prime	PASS
$509 \equiv 1 \pmod 4$	PASS
Miyamoto-for-508 ineligible	PASS
Paley I/II fail for 508	PASS
Reject all-ones 4×4	PASS
SDS-127 spin constant λ	PASS
SDS-127 $\sum k_i = n + \lambda$	PASS
Paley conference order 6	PASS
Miyamoto H_{20} from H_4	PASS

All 15/15 unit tests PASS. End-to-end H_{508} and H_{2036} verifications also PASS.

7. Non-claims

- Full Hadamard conjecture: **OPEN** (Band-3).
- No Clay Millennium problem closed; `clay_closed = false`.
- GATE-04 collection: **LOCKED**.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Orders beyond 2036 (e.g. other Miyamoto-eligible gaps 2116, 2164, ...) are **not** claimed constructed here.

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_508_to_2036_miyamoto
python3 verify_hadamard_508_2036.py
```

Emits `results.json`, `literature_ledger.json`, `SUMMARY.txt`, and on success `H508_pm1.npz`, `H2036_pm1.npz`.

9. Conclusion

Outcome. PASS — H_{508} and H_{2036} constructed and verified on box. Sage was absent; construction used published SDS data + NumPy. Full Hadamard conjecture remains **OPEN**.

Next optional. Triage/construct other post-2000 Miyamoto-eligible gaps (2116, 2164, 2228, 2276) contingent on matching H_{q-1}; or Track B privacy / Object 1 meter await.

Shelf. `2026-09-14_Band1_Hadamard_508_to_2036_Miyamoto.md` / `.pdf` under `/workspace/DOWNLOAD_PDFs/`.